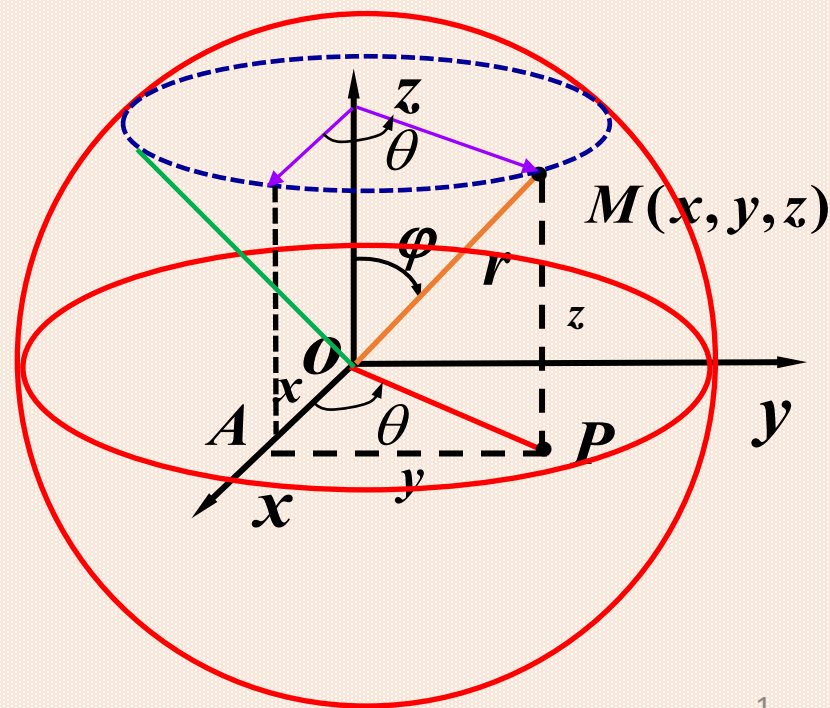


2.4、利用球坐标计算三重积分

设 $M(x, y, z)$ 为空间内一点，则点 M 可用三个有次序的数 r, φ, θ 来确定，其中 r 为原点 O 与点 M 间的距离， φ 为有向线段 OM 与 z 轴正向所夹的角， θ 为从正 z 轴来看自 x 轴按逆时针方向转到有向线段 OP 的角，这里 P 为点 M 在 xoy 面上的投影，这样的三个数 r, φ, θ 就叫做点 M 的球坐标。

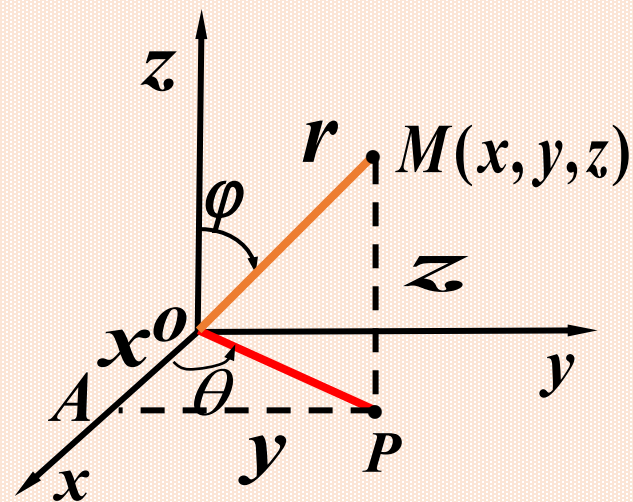


设点 P 在 x 轴上的投影为 A ,

则 $OA = x$, $AP = y$, $PM = z$.

球坐标与直角坐标的关系为

$$\begin{cases} x = r \sin \varphi \cos \theta, \\ y = r \sin \varphi \sin \theta, \\ z = r \cos \varphi. \end{cases}$$



规定: $0 \leq r < +\infty$,

$0 \leq \varphi \leq \pi$,

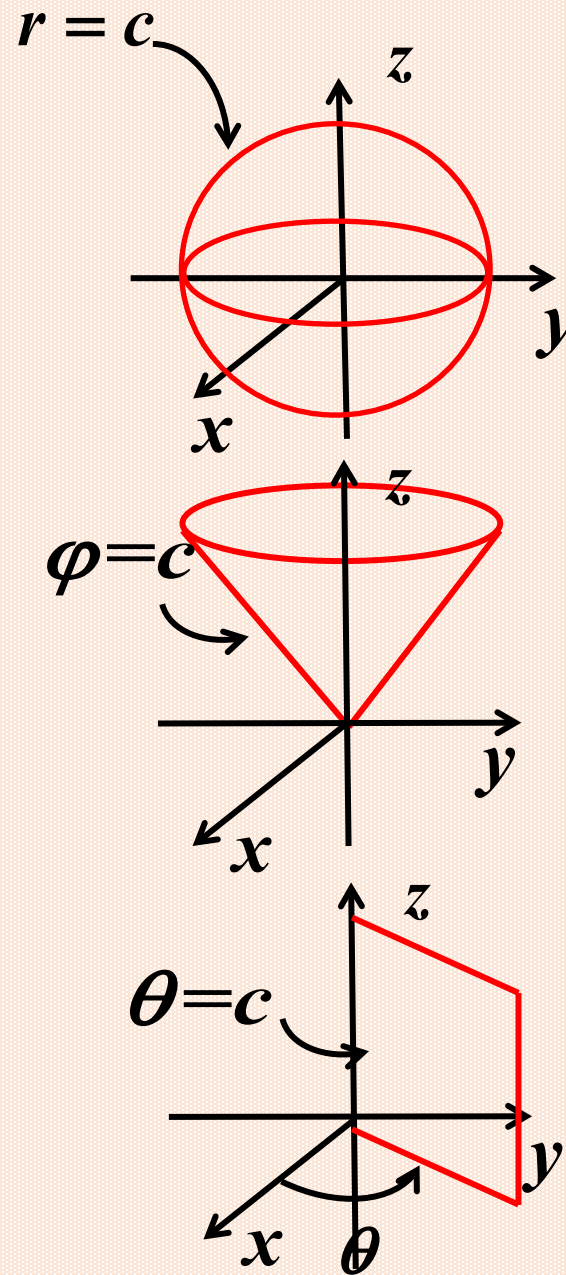
$0 \leq \theta \leq 2\pi$.

如图，三坐标面分别为

r 为常数 $\Rightarrow$ 球 面;

φ 为常数 $\Rightarrow$ 圆锥面;

θ 为常数 $\Rightarrow$ 半平面.



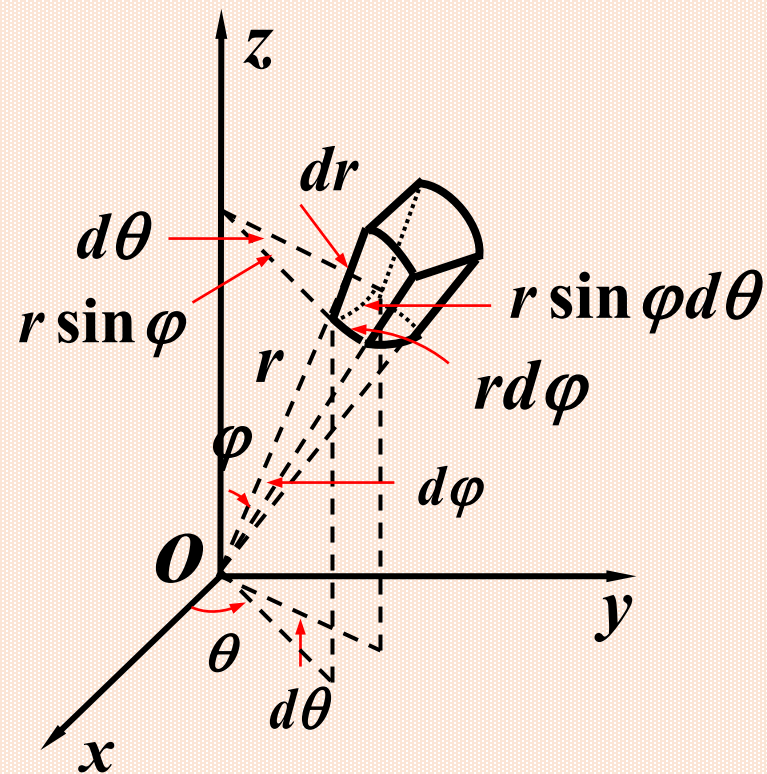
如图,

球坐标系中的体积元素为

$$dV = r^2 \sin \varphi dr d\varphi d\theta,$$

$$\iiint_{\Omega} f(x, y, z) dx dy dz =$$

$$\iiint_{\Omega} f(r \sin \varphi \cos \theta, r \sin \varphi \sin \theta, r \cos \varphi) r^2 \sin \varphi dr d\varphi d\theta.$$



例1 计算 $\iiint_{\Omega} (x^2 + y^2 + z^2) dV$. 其中 Ω 由曲面

$$z = \sqrt{x^2 + y^2} \text{ 和 } x^2 + y^2 + z^2 = R^2 \text{ 围成 } (R > 0)$$

解 将 Ω 向 xoy 面投影, 得

$$0 \leq \theta \leq 2\pi.$$

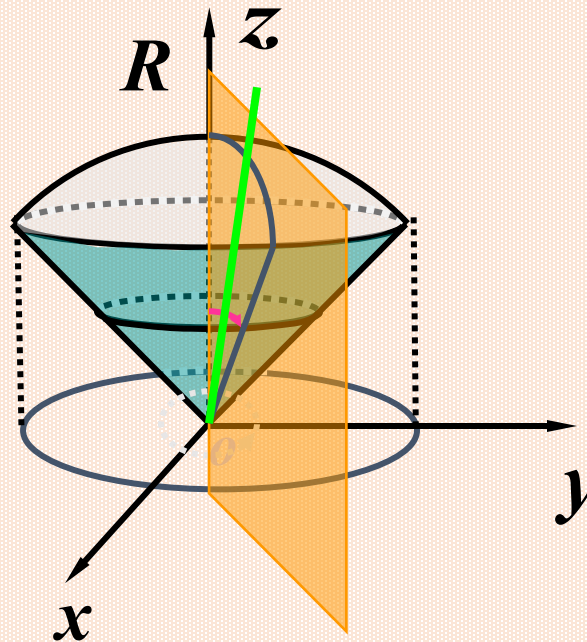
任取一 $\theta \in [0, 2\pi]$, 过 z

轴作半平面, 得

$$0 \leq \varphi \leq \frac{\pi}{4}.$$

在半平面上, 任取一 $\varphi \in [0, \frac{\pi}{4}]$,

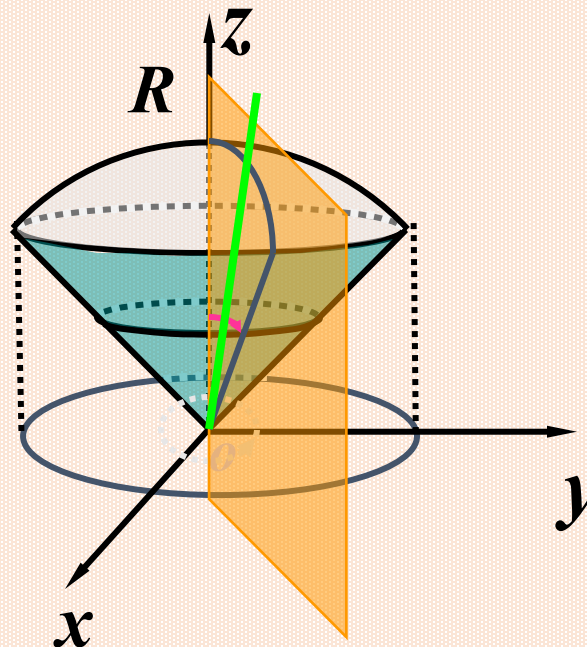
过原点作射线, 得 $0 \leq r \leq R$.



即

$$\Omega: \begin{cases} 0 \leq \theta \leq 2\pi, \\ 0 \leq \varphi \leq \frac{\pi}{4}, \\ 0 \leq r \leq R. \end{cases}$$

$$\begin{aligned} & \iiint_{\Omega} (x^2 + y^2 + z^2) dV \\ &= \iiint_{\Omega} r^2 \cdot r^2 \sin \varphi dr d\varphi d\theta \\ &= \int_0^{2\pi} d\theta \int_0^{\pi/4} d\varphi \int_0^R r^4 \sin \varphi dr \\ &= \frac{1}{5} \pi R^5 (2 - \sqrt{2}). \end{aligned}$$



$$\begin{cases} x = r \sin \varphi \cos \theta, \\ y = r \sin \varphi \sin \theta, \\ z = r \cos \varphi. \end{cases}$$

$$dV = r^2 \sin \varphi dr d\varphi d\theta$$

例2 计算 $\iiint_{\Omega} (x+z)dV$, 其中 Ω 由 $z = \sqrt{x^2 + y^2}$ 与 $z = \sqrt{1-x^2-y^2}$ 所围成的.

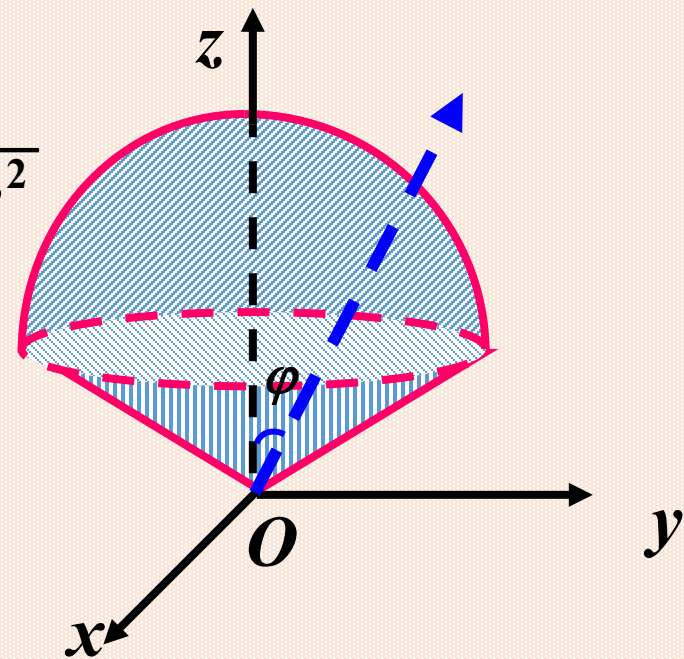
解 $\because \Omega$ 关于 $yo z$ 面为对称,

$f(x, y, z) = x$ 为 x 的奇函数,

故有 $\iiint_{\Omega} x dV = 0$.

$$\therefore \iiint_{\Omega} (x+z)dV = \iiint_{\Omega} z dV \text{ (利用球面坐标)}$$

$$= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} d\varphi \int_0^1 r \cos \varphi \cdot r^2 \sin \varphi dr = \frac{\pi}{8}.$$



$$z = \sqrt{1-x^2-y^2} \Rightarrow r = 1,$$

$$z = \sqrt{x^2+y^2} \Rightarrow \varphi = \frac{\pi}{4},$$

$$\Omega : 0 \leq r \leq 1, 0 \leq \varphi \leq \frac{\pi}{4},$$

$$0 \leq \theta \leq 2\pi$$

例3 求曲面 $x^2 + y^2 + z^2 \leq 2a^2$ 与 $z \geq \sqrt{x^2 + y^2}$ 所围成的立体体积.

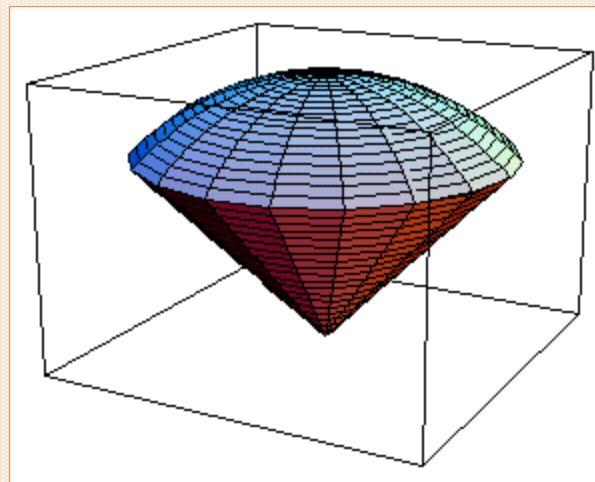
解 Ω 由锥面和球面围成, 采用球坐标,

$$\text{由 } x^2 + y^2 + z^2 = 2a^2$$

$$\Rightarrow r = \sqrt{2}a,$$

$$z = \sqrt{x^2 + y^2} \Rightarrow \varphi = \frac{\pi}{4},$$

$$\Omega: \quad 0 \leq \theta \leq 2\pi, \quad 0 \leq \varphi \leq \frac{\pi}{4}, \quad 0 \leq r \leq \sqrt{2}a,$$



由三重积分的性质知 $V = \iiint_{\Omega} dx dy dz,$

$$V = \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} d\varphi \int_0^{\sqrt{2}a} r^2 \sin \varphi dr$$

$$= 2\pi \int_0^{\frac{\pi}{4}} \sin \varphi \cdot \frac{(\sqrt{2}a)^3}{3} d\varphi = \frac{4}{3}\pi(\sqrt{2}-1)a^3.$$

例4 计算三重积分 $\iiint_{\Omega} \frac{\cos \sqrt{x^2 + y^2 + z^2}}{\sqrt{x^2 + y^2 + z^2}} dV$.其中

$$\Omega = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 4\pi^2, x^2 + y^2 + z^2 \geq \pi^2\}.$$

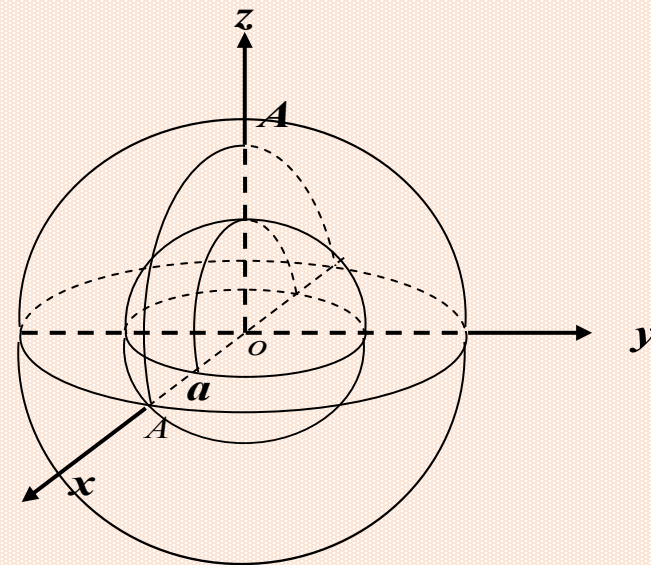
解 积分区域 Ω 的图形如图.

在球面坐标系下

$$\Omega: 0 \leq \theta \leq 2\pi, 0 \leq \varphi \leq \pi, \pi \leq r \leq 2\pi$$

$$\text{故有 } \iiint_{\Omega} \frac{\cos \sqrt{x^2 + y^2 + z^2}}{\sqrt{x^2 + y^2 + z^2}} dV = \int_0^{2\pi} d\theta \int_0^{\pi} d\varphi \int_{\pi}^{2\pi} \frac{\cos r}{r} \cdot r^2 \sin \varphi dr$$

$$= 4\pi \int_{\pi}^{2\pi} r \cos r dr = 4\pi (r \sin r + \cos r) \Big|_{\pi}^{2\pi} = 8\pi.$$



例5 计算三重积分 $\iiint_{\Omega} (x^2 + y^2) dV$. 其中 Ω 是由球面 $z = \sqrt{A^2 - x^2 - y^2}$

$z = \sqrt{a^2 - x^2 - y^2} (A > a > 0)$ 和平面 $z = 0$ 所确定的闭区域.

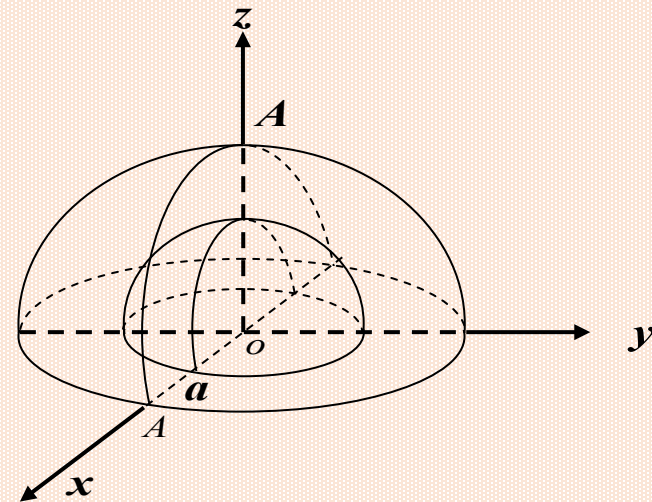
解 积分区域 Ω 的图形如图.

在球面坐标系下

$$\Omega: 0 \leq \theta \leq 2\pi, 0 \leq \varphi \leq \frac{\pi}{2}, a \leq r \leq A$$

$$\text{故有 } \iiint_{\Omega} (x^2 + y^2) dV = \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_a^A r^2 \sin^2 \varphi \cdot r^2 \sin \varphi dr$$

$$= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} \sin^3 \varphi d\varphi \int_a^A r^4 dr = 2\pi \cdot \frac{2}{3} \cdot \frac{1}{5} (A^5 - a^5) = \frac{4}{15} \pi (A^5 - a^5)$$



例6 计算 $I = \iiint_{\Omega} (x^2 + y^2 + z^2) dx dy dz$ 其中 Ω 是由球面

$x^2 + y^2 + z^2 = z$ 所限定的球域.

解 采用球坐标

$$x^2 + y^2 + z^2 = z \rightarrow r = \cos \varphi$$

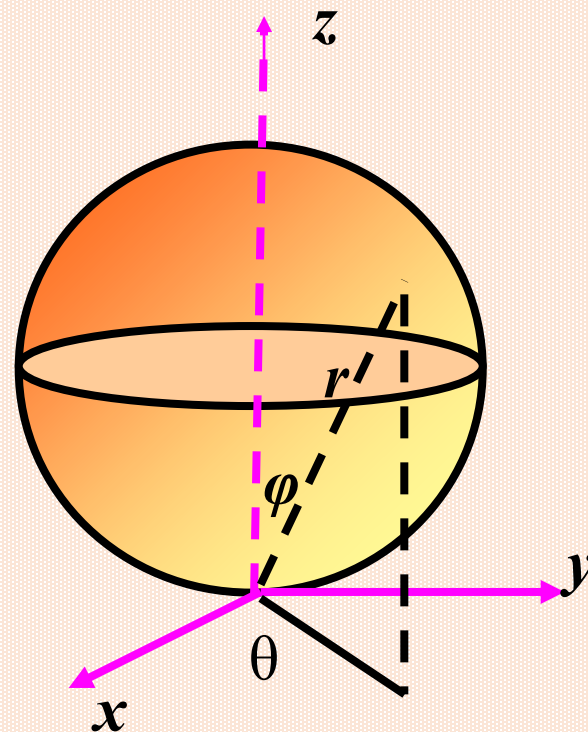
$$\therefore \Omega: 0 \leq \theta \leq 2\pi, 0 \leq \varphi \leq \frac{\pi}{2},$$

$$0 \leq r \leq \cos \varphi,$$

$$\text{故有 } I = \iiint_{\Omega} (x^2 + y^2 + z^2) dx dy dz$$

$$= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_0^{\cos \varphi} r^2 \cdot r^2 \sin \varphi dr = 2\pi \cdot \frac{1}{5} \int_0^{\frac{\pi}{2}} \cos^5 \varphi \sin \varphi d\varphi$$

$$= \frac{2\pi}{5} \left(-\frac{1}{6} \cos^6 \varphi \right) \bigg|_0^{\frac{\pi}{2}} = \frac{\pi}{15}.$$



例7 计算三重积分 $\iiint_{\Omega} x^2 dV$

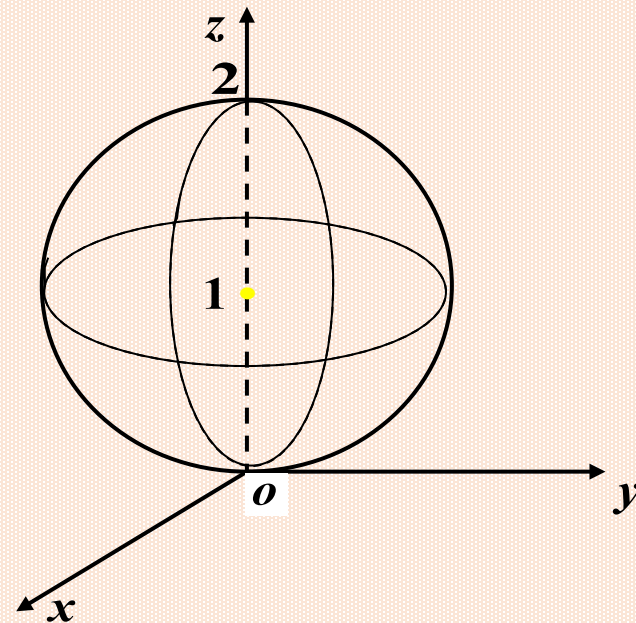
其中 Ω 是球形闭区域: $x^2+y^2+z^2\leq 2z$.

解 采用球坐标

$$x^2 + y^2 + z^2 = 2z \rightarrow r = 2 \cos \varphi$$

$$\therefore \Omega: 0 \leq \theta \leq 2\pi, 0 \leq \varphi \leq \frac{\pi}{2},$$

$$0 \leq r \leq 2 \cos \varphi,$$



$$\begin{aligned}
\iiint_{\Omega} x^2 dV &= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_0^{2\cos\varphi} r^4 \sin^3 \varphi \cos^2 \theta dr \\
&= \int_0^{2\pi} \cos^2 \theta d\theta \int_0^{\frac{\pi}{2}} \frac{32}{5} \sin^3 \varphi \cos^5 \varphi d\varphi \\
&= \frac{32}{5} \int_0^{2\pi} \frac{1+\cos 2\theta}{2} d\theta \int_0^{\frac{\pi}{2}} -(1-\cos^2 \varphi) \cos^5 \varphi d\cos \varphi \\
&= \frac{32\pi}{5} \left[-\frac{1}{6} \cos^6 \varphi + \frac{1}{8} \cos^8 \varphi \right]_0^{\frac{\pi}{2}} = \frac{32\pi}{5} \frac{1}{24} = \frac{4\pi}{15}
\end{aligned}$$

或者 由于 Ω 关于平面 $y=x$ 对称,

$$\begin{aligned}
\iiint_{\Omega} x^2 dV &= \iiint_{\Omega} y^2 dV = \frac{1}{2} \iiint_{\Omega} (x^2 + y^2) dV \\
&= \frac{1}{2} \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_0^{2\cos\varphi} r^4 \sin^3 \varphi dr = \pi \int_0^{\frac{\pi}{2}} \frac{32}{5} \sin^3 \varphi \cos^5 \varphi d\varphi \cdots = \frac{4\pi}{15}
\end{aligned}$$

计算 $I = \iiint_{\Omega} \sqrt{x^2 + y^2 + z^2} dV$

其中 Ω 由 $x^2 + y^2 + (z-1)^2 = 1$ 围成.

$I =$ [填空1]

正常使用填空题需3.0以上版本雨课堂

作答

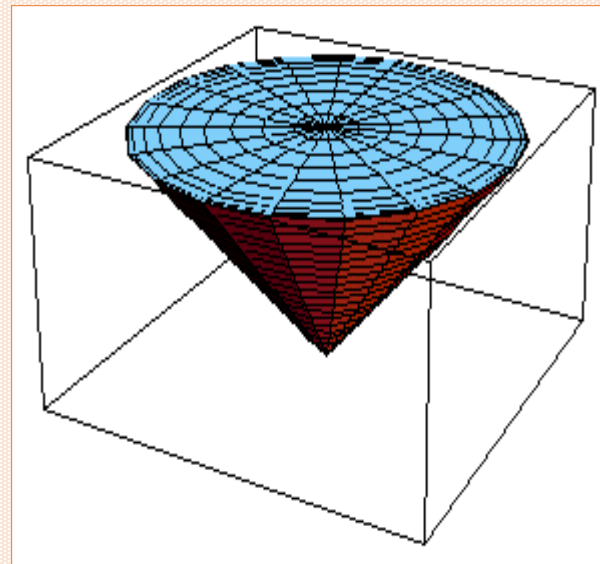
例8 计算 $I = \iiint_{\Omega} (x^2 + y^2) dx dy dz$, 其中 Ω 是锥面 $x^2 + y^2 = z^2$, 与平面 $z = a$ ($a > 0$) 所围的立体.

解1 采用球坐标

$$\because z = a \Rightarrow r = \frac{a}{\cos \varphi},$$

$$x^2 + y^2 = z^2 \Rightarrow \varphi = \frac{\pi}{4},$$

$$\therefore \Omega: 0 \leq \theta \leq 2\pi, \quad 0 \leq \varphi \leq \frac{\pi}{4}, \quad 0 \leq r \leq \frac{a}{\cos \varphi},$$



$$I = \iiint_{\Omega} (x^2 + y^2) dx dy dz$$

$$= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} d\varphi \int_0^{\frac{a}{\cos \varphi}} r^4 \sin^3 \varphi dr$$

$$= 2\pi \int_0^{\frac{\pi}{4}} \sin^3 \varphi \cdot \frac{1}{5} \left(\frac{a^5}{\cos^5 \varphi} - 0 \right) d\varphi$$

$$= \frac{\pi}{10} a^5.$$

解2 采用柱坐标

$$\because x^2 + y^2 = z^2 \Rightarrow z = r, \quad D: x^2 + y^2 \leq a^2,$$

$$\Omega: 0 \leq \theta \leq 2\pi, \quad 0 \leq r \leq a, \quad r \leq z \leq a$$

$$I = \iiint_{\Omega} (x^2 + y^2) dx dy dz = \int_0^{2\pi} d\theta \int_0^a r dr \int_r^a r^2 dz$$

$$= 2\pi \int_0^a r^3 (a - r) dr = 2\pi \left[a \cdot \frac{a^4}{4} - \frac{a^5}{5} \right] = \frac{\pi}{10} a^5.$$

例9 计算 $\iiint_{\Omega} (x+z)dV$, 其中 Ω 由 $z = \sqrt{x^2 + y^2}$ 与

$z = 1 + \sqrt{1 - x^2 - y^2}$ 所围成的.

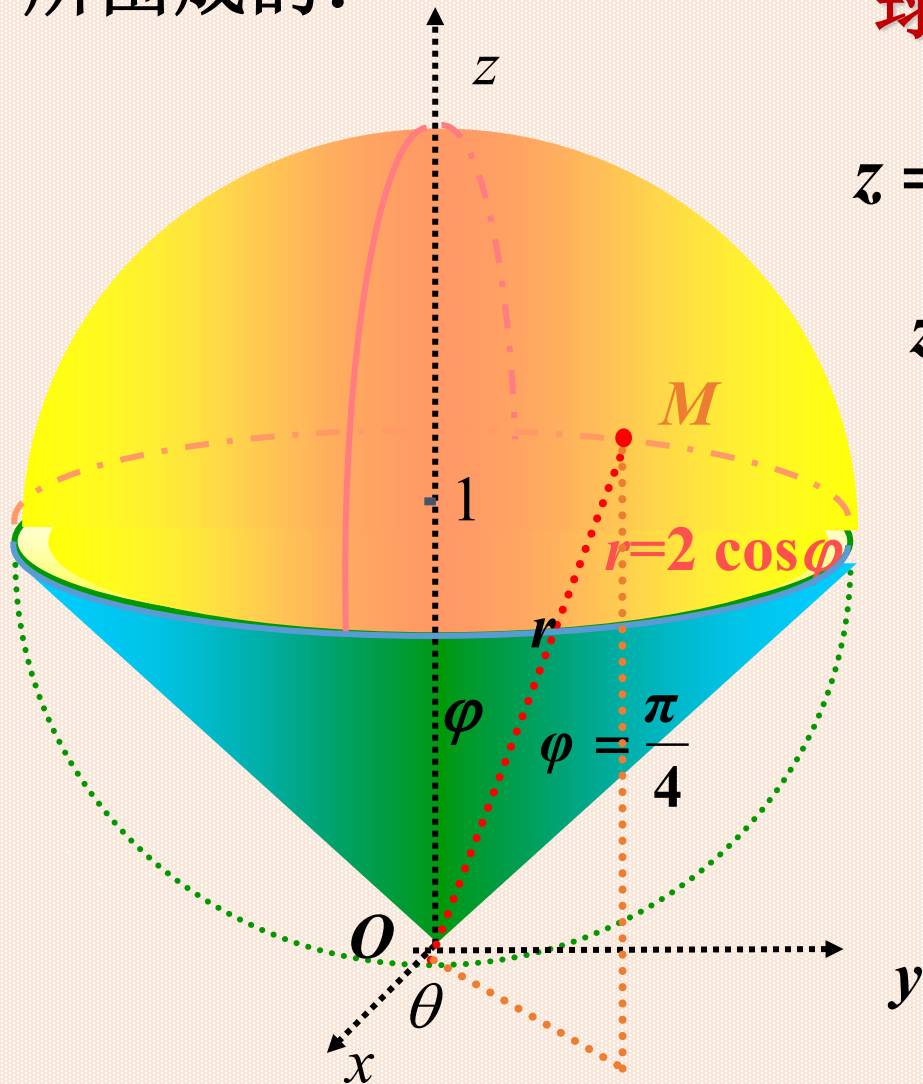
由
$$\begin{cases} z = \sqrt{x^2 + y^2} \\ z = 1 + \sqrt{1 - x^2 - y^2} \end{cases}$$

消去 z 得 $x^2 + y^2 = 1$

于是交线方程为

$$\begin{cases} x^2 + y^2 = 1 \\ z = 1 \end{cases}$$

直角坐标, 柱坐标



球坐标下

$$z = \sqrt{x^2 + y^2} \Rightarrow \varphi = \frac{\pi}{4}$$

$$z = 1 + \sqrt{1 - x^2 - y^2}$$

$$\Rightarrow r = 2 \cos \varphi$$

故在球坐标下

$$\Omega: 0 \leq \theta \leq 2\pi$$

$$0 \leq \varphi \leq \frac{\pi}{4}$$

$$0 \leq r \leq 2 \cos \varphi$$

解 $\because \Omega$ 关于 $yo z$ 面为对称, $f(x, y, z) = x$ 为 x 的奇函数, 有 $\iiint_{\Omega} x dV = 0$.

$$\therefore \iiint_{\Omega} (x + z) dV = \iiint_{\Omega} z dV$$

在球坐标下

$$\Omega = \left\{ (\theta, \varphi, r) \mid 0 \leq \theta \leq 2\pi, 0 \leq \varphi \leq \frac{\pi}{4}, 0 \leq r \leq 2 \cos \varphi \right\}$$

于是

$$\iiint_{\Omega} z dV = \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} d\varphi \int_0^{2\cos\varphi} r \cos \varphi \cdot r^2 \sin \varphi dr = \frac{7\pi}{6}.$$

例10 利用对称性简化计算

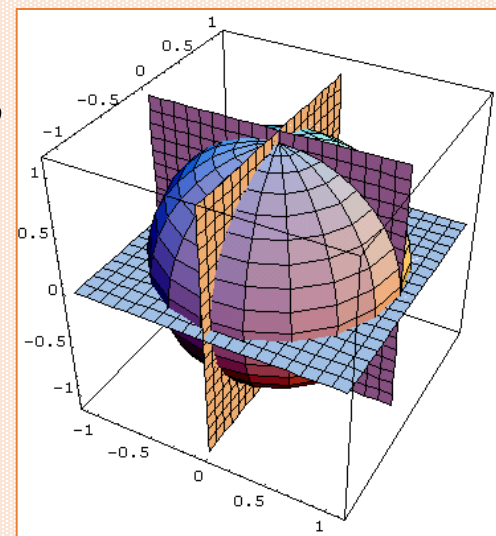
$$\iiint_{\Omega} \frac{z \ln(x^2 + y^2 + z^2 + 1)}{x^2 + y^2 + z^2 + 1} dx dy dz$$

其中积分区域 $\Omega = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 1\}$.

解 积分域关于三个坐标面都对称,

被积函数是 z 的奇函数,

$$\iiint_{\Omega} \frac{z \ln(x^2 + y^2 + z^2 + 1)}{x^2 + y^2 + z^2 + 1} dx dy dz = 0.$$



例11 计算 $I = \iiint_{x^2+y^2+z^2 \leq a^2} (lx^2 + my^2 + nz^2) dx dy dz$

解 由轮换对称性

$$\begin{aligned} \iiint_{x^2+y^2+z^2 \leq a^2} x^2 dx dy dz &= \iiint_{x^2+y^2+z^2 \leq a^2} y^2 dx dy dz = \iiint_{x^2+y^2+z^2 \leq a^2} z^2 dx dy dz \\ &= \frac{1}{3} \iiint_{x^2+y^2+z^2 \leq a^2} (x^2 + y^2 + z^2) dx dy dz \end{aligned}$$

而 $\iiint_{x^2+y^2+z^2 \leq a^2} (x^2 + y^2 + z^2) dx dy dz = \int_0^{2\pi} d\theta \int_0^\pi d\varphi \int_0^a r^2 \cdot r^2 \sin \varphi dr = \frac{4}{5} \pi a^5$

故 $I = \frac{l+m+n}{3} \iiint_{x^2+y^2+z^2 \leq a^2} (x^2 + y^2 + z^2) dx dy dz = \frac{4}{15} (l+m+n) \pi a^5$

练习 计算 $I = \iiint_{x^2+y^2+z^2 \leq a^2} (lx + my + nz + d)^2 dx dy dz$

解题方法流程图

